

FindIt System Manual

Version 0.5.15 (Manual)

Build Date: January 30, 2026

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FindIt: A LoRaWAN-Based Asset Tracking System

by

Ryan Davis, Cory Vitanza, and Roy Tas

Group 2

Stevens Institute of Technology

Fall 2025

Version 0.5.15 (Manual)

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Table 1: Document Update History

Date	Updates
1/30/2026	RD, RT, CV: • Updated Weekly Report 1 for the week of 1/30/26. • Updated MVC React UML Class diagram.
1/20/2026	RD, RT, CV: • Updated Development Notes for sensecap LoRa configurations for LNS.
12/17/2025	RD, RT, CV: • Added Preliminary Implementation, providing an overview of our prototype demo. • Fixed links between requirements and use cases in Use Cases Chapter, for dynamic requirement labeling and references between chapters. • Updated to Version 0.4.13
12/12/2025	RD, RT, CV: • Added Weekly Report 13 to appendix
12/05/2025	RD, RT, CV: • Added Weekly Report 12 to appendix. • Updated Development Notes for Development board configurations
11/18/2025	RD, RT, CV: • UML Class diagrams for our Logical view, a package diagram for our development view, an activity diagram for our process view, and a component diagram for our physical view. These are necessary components of the 4+1 view. • Added labels to Requirements tables, allowing for simpler naming and references of requirements.

Table 1: Document Update History

Date	Updates
11/14/2025	RD, RT, CV: - Added User Interface Design, Logical View, Development View, Process View, and Physical View chapters, representing the 4+1 view of the FindIt system. - Added manual versioning for documentation (for now). - Added Figma Paper Prototype example for the FindIt user interface
11/07/2025	RD, RT, CV: - Added Weekly Report 10 - Dynamically referenced requirements in Requirements Chapter.
10/31/2025	RD, RT, CV: - Added Weekly Report 9 - Added more activity diagrams to replicate the flows of our use cases
10/24/2025	RD, RT, CV: - Added Development Notes to appendix to track and log any specific processes or findings that we may come across throughout our development. - Mapped Requirements to use cases, and added references to user stories. - Added Weekly Report 8
10/17/2025	RD, RT, CV: - Added Use Cases Chapter, including: - Table of Use Cases with all six primary use cases. - Use Case Diagram. - Detailed Use Case Descriptions for each primary use case. - System Activity Diagram - Added Weekly Report 7
10/10/2025	RD, RT, CV: - Added Requirements Chapter - Added User Stories Chapter. - Added Use Cases Chapter. - Updated Glossary - Added Weekly Report 6
10/02/2025	RD, RT, CV: Added Development Plan chapter.
09/30/2025	RD, RT, CV: - Added Bibliography to appendix - Moved Weekly Reports to appendix.
09/25/2025	RD, RT, CV: Added Weekly Report 4
09/18/2025	RD, RT, CV: - Edited and Finished Team Declaration chapter - Added Weekly Report 3

Table 1: Document Update History

Date	Updates
09/16/2025	RD, RT, CV: Added Team Declaration chapter
09/12/2025	RD, RT, CV: - Added Roy Tas's introduction - Added Weekly Report 2 - Created System overview diagram
09/04/2025	RD, RT, CV: - Added introduction page, introducing the team. - Added Weekly Report 1

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Chapter 1

Introduction

- Ryan Davis, Cory Vitanza, and Roy Tas

My name is Ryan Davis, and I am a Software Engineering major. I am in my senior year of study pursuing my bachelor's degree. I hope to continue my Software Engineering career one day as the founder of a startup company.

Hi, my name is Cory Vitanza, a 4th-year Software Engineering student also currently pursuing a Master's in Engineering Management. I've gained coding experience from my classes and real-world experience in QA and testing during my internship at Trimble MAPS. After my time here at Stevens, I plan to continue working in QA, hopefully in a management position soon enough. My favorite coding language is C++, and I love working on challenging projects. Outside of coding, I'm into music, cars, video games, and watching / playing sports, especially basketball and football

My name is Roy Tas I am a senior Engineering Management major concentrating in Financial Engineering. I have some C++ and Python experience but I will be focusing more on the scalability and logistics of our project. Post-graduation I want to pursue a career in jewelry manufacturing.

Chapter 2

Team Declaration

- Ryan Davis, Cory Vitanza, and Roy Tas

2.1 Team Information

- **Team Name:** FindIt

- **Team Members:**

1. Ryan Davis
2. Cory Vitanza
3. Roy Tas

2.2 Project Proposal

Patrons with disabilities such as autism or conditions such as dementia tend to wander or get lost, causing families and caregivers of such patrons great amounts of stress. We are FindIt and we aim to mitigate stress, with our new tracking solution employing [LoRaWAN \(Long Range Wide Area Network\)](#) technology to track locations at far distances, with extensive battery life.

Our device aims to demonstrate that a low-cost and accessible solution can balance affordability, reliability, and ease of use- as compared to current tracking solutions like AirTags or other commercial tracking devices. We envision FindIt as a scalable option that extends beyond families to institutional applications such as nursing homes, group homes, and schools, where safety and monitoring are critical. Our device comforts families with efficient tracking tools to ensure comfort when their loved ones are lost.

2.3 Mission Statement

The mission of FindIt is to design and deliver a reliable, affordable, and user-friendly location tracking solution for families and caregivers of individuals at risk of wandering. Our goal is to reduce stress by providing a reliable tool that allows quick and efficient tracking in critical situations. The business need is the lack of affordable long-range tracking technology for families, and the business intent is to bridge that gap by leveraging LoRaWAN for extended battery life and coverage.

2.4 Key Drivers

Several key drivers motivate the development of this solution.

There is a great need for improved tracking solutions for wandering patrons. Wandering is prevalent among such groups with autism or dementia, and existing solutions are expensive, insufficient, and at times inaccurate. We aim to minimize these issues with the development of our device.

Current solutions for asset tracking, like AirTags or cellular-based trackers, can become expensive for long-term usage. Using LoRaWAN protocols, our device reduces recurring costs and

enhances battery life.

Technological accessibility is also an important motivator, as our group aims to reduce the stress of figuring out an application when time is of the essence. Our service is also subscription-free, in order to benefit all those in need.

LoRaWAN's versatility allows for uses beyond families. Group facilities such as schools, assisted living centers, or hospitals could benefit from a low-cost, universal system to monitor the safety of many individuals simultaneously.

2.5 Key Constraints

- **Network Coverage:** LoRaWAN's effectiveness depends on gateway proximity. Limited coverage in some areas may impact reliability, requiring design options for sparse networks. LoRaWAN coverage maps detail that Urban areas, specifically in the Northeast, have excellent coverage.
- **Battery Life vs. Device Size:** Maintaining strong battery life while preserving the small size of this device is optimal for the constant carriage of our unit. The device must remain lightweight and comfortable for continuous use.
- **Privacy and Security:** Storing and sending location data leads to privacy concerns. Secure data handling, LoRaWAN encryption, and user-controlled permissions are necessary to maintain trust with the user.
- **Cost Targets:** Affordability is a primary selling point. Hardware, manufacturing, and deployment costs must remain competitive without sacrificing reliability.
- **Interface Simplicity:** Achieving a clean, intuitive interface while maintaining accessibility features is critical for the tracking unit to reach diverse user bases with differing technologi-

cal expertise.

Chapter 3

Development Plan

- Ryan Davis, Cory Vitanza, and Roy Tas

3.1 Introduction

The purpose of this project is to design, implement, and validate a location-tracking and data communication system using LoRaWAN-enabled hardware integrated with a cloud-based software platform. The system aims to provide reliable long-range wireless data transmission between embedded devices and a backend service, with supporting web-based tools for monitoring and management. By combining embedded development, cloud infrastructure, and a user-facing interface, this project demonstrates the end-to-end development process of a distributed system.

Success criteria for the project include (1) establishing reliable LoRaWAN communication between the embedded board and backend, (2) ensuring data is properly stored, processed, and displayed through the frontend application, and (3) passing acceptance testing with the Wearable Robotics Systems Lab. From a user perspective, the system enables simple, reliable collection and visualization of data in real time, serving as a proof-of-concept platform that can be extended to future applications.

3.2 Roles and Responsibilities

Clear definition of roles and responsibilities ensures accountability and effective collaboration within the team. Since this project is being developed by a small group, several members will serve multiple roles across development, testing, documentation, and risk management. The following assignments identify primary responsibilities, while recognizing that all team members will contribute collaboratively to design decisions, implementation, and review processes.

1. Development Lead: Ryan
2. Buildmeister: Roy
3. Architect: Cory
4. Developers: Ryan (Backend), Cory (Frontend), Roy (Infrastructure)
5. Test Lead: Cory
6. Testers: All
7. Documentation: All
8. Documentation Editor: All
9. Designer: Cory
10. User advocate: Ryan Davis
11. Risk Management: Roy
12. System Administrator: Ryan
13. Modification Request Board: All
14. Requirements Resource: Roy

15. Customer Representative: All
16. Customer responsible for acceptance testing: Wearable Robotics Systems Lab

3.3 Method

These are unique to software development, although there may be some overlap.

3.3.1 Software

1. Language(s):

- C (Arduino IDE)
- Python (3.x)
- TypeScript (React)
- SQL (PostgreSQL via Supabase)

2. Operating System(s):

- Arduino embedded OS (board-specific)
- Linux/Ubuntu (for Docker and AWS deployment)
- macOS/Windows (development environments)

3. Software packages/libraries used:

- LoRaWAN protocol (TCP/IP stack integration)
- Supabase (PostgreSQL backend service)
- Google Maps API
- React (TypeScript frontend framework)

- Docker (containerization)
- AWS (deployment and hosting)
- GitHub (version control)
- Kanban Board (Agile task management)

4. Code conventions:

- GitHub repository guidelines for commits and pull requests
- ESLint + Prettier for TypeScript (frontend)
- PEP 8 for Python (backend tools)
- Arduino C style conventions

3.3.2 Hardware

1. Development Hardware

- Wio Tracker 1110 Development Board
- Semtech LR1110 LoRa® transceiver

2. Test Hardware

- Wio Tracker 1110 Development Board
- LoRaWAN Gateway (connected to The Things Stack)

3. Target/Deployment Hardware

- Wio Tracker 1110 Development Board
- Semtech LR1110 LoRa® transceiver
- LoRaWAN Gateway (The Things Stack integration for live deployment)

3.3.3 Backup plan (individual and project)

In case the Wio Tracker is not capable of the deployment of our software, the FindIt team will need to research and develop board alternatives. One option for this scenario would be to use a Raspberry Pi Nano, implemented with an external LoRaWAN module to allow for data transmission. Even with such addition, it is necessary for our unit to maintain battery consumption amongst boards, ensuring our unit has extensive battery life for users.

3.3.4 Review Process

1. Will you do architecture, usability, design, security, privacy or code reviews?
 - Code reviews will be completed upon the completion of Agile tasks on our Kanban board.
2. What approach will you use for the reviews (formal, informal, corporate standard)?
 - Informal reviews will be employed to achieve the approval of all members of the team, ensuring that all cases are handled in new implementation.
3. Who is responsible for the reviews and resolving any issues uncovered by the reviews?
 - Upon the implementation of a component, the other two members of the team will perform a code review. Any issue uncovered from code review will be resolved by the member working on the component.

3.3.5 Build Plan

1. Revision control system and repository used
 - GitHub (Version Control)

2. Continuous integration

- Circle CI

3. Deadlines for the builds

- Depending on our planned sprints, our Kanban board will present deadlines based on tasks.

4. Regression test process – see test plan

3.3.6 Modification Request Process

1. MR tool: For modification requests, we will use our Kanban board on GitHub to issue any modification requests as a task in the board.
2. Decision process: Decisions for modification will be decided during heartbeat meetings, after a discussion among the team concerning the modification. The team will proceed to assign a task to the Kanban board on the methodology and developers necessary for the task.
3. State whether there will be two process streams one during development and one after development:
 - During development, there will be two process streams. One is this programming of the development board and the other will be for the front-end user interface. The two streams will be integrated for deployment.

3.4 Virtual and Real Workspace

3.4.1 Virtual Workspace

Our team will utilize texts for direct communication, as well as a Discord server used for the sharing of files, resources, and communication. GitHub will also be used for secondary communication, implementing comments and changes to the code base via commits and merge requests.

3.4.2 Real Workspace

The team's regular workspace will be the Software Engineering Lab, primarily for heartbeat meetings and the development of our program. Meetings with Dr. Zanotto will be held in his office or over Zoom, when necessary.

3.5 COMMUNICATION PLAN

3.5.1 Heartbeat Meetings

Heartbeat meetings will be held at least twice per week with all group members present. These meetings serve to take the “pulse” of the project by providing quick updates, raising blockers, and reviewing open issues and risks. Each meeting will follow a structured agenda that includes:

- Individual progress updates
- Discussion of current blockers or risks
- Review of action items and open issues

Notes from each meeting will be recorded in the project document, and the GitHub Kanban board will be updated accordingly. These meetings are designed to be short, ideally during our allotted class-time.

3.5.2 Status Meetings

Status meetings will occur once a month and will include the project group and Dr. Damiano Zanno. The purpose of these meetings is to present overall project progress, demonstrate completed work, and receive feedback on areas for improvement. If issues arise during these meetings, they will be addressed separately in an issues meeting. These meetings are intended to be concise, focusing on high-level updates and progress checkpoints.

3.5.3 Issues Meetings

Issues meetings will be scheduled whenever significant problems arise that require group consensus or immediate attention. Alerts for these meetings will typically be raised through a Discord message. If consensus is reached that a meeting is necessary, it will be scheduled as soon as all group members are available. These meetings provide a dedicated space for problem-solving and decision-making outside of the regular heartbeat meetings.

3.6 Timeline and Milestones

The following milestones outline the major deliverables for the project. Each milestone represents a 100% complete item, with critical participants, begin and end dates, and deliverables clearly defined.

- **Milestone 1: Project Setup & Requirements Finalization**

- Begin: Oct 3, 2025 End: Oct 17, 2025
- Deliverables: Project repository initialized, Kanban board configured, requirements document finalized, hardware/software resources received and confirmed.

- **Milestone 2: Core System Architecture & Prototype Communication**

- Begin: Oct 17, 2025 End: Oct 31, 2025
- Deliverables: Draft system architecture, initial Arduino board programming for basic LoRaWAN communication, mock frontend interface, database schema in Supabase.
- **Milestone 3: Integration of Hardware and Software Components**
 - Begin: Oct 31, 2025 End: Nov 14, 2025
 - Deliverables: Hardware successfully transmitting data to backend, frontend connected to database, Docker setup for local deployment, integration testing initiated.
- **Milestone 4: Feature Implementation & System Testing**
 - Begin: Nov 14, 2025 End: Nov 28, 2025
 - Deliverables: Full set of planned features implemented, CI/CD pipeline operational, unit + integration tests passing, preliminary user documentation drafted.
- **Milestone 5: Final Review & Delivery**
 - Begin: Nov 28, 2025 End: Dec 12, 2025
 - Deliverables: Final acceptance testing completed, documentation finalized, project demonstration delivered, final repository and deployment package prepared for submission.

This document will be re-issued at each milestone to reflect progress and highlight changes since the previous revision.

3.7 Testing Policy/Plan

Our testing policy combines automated and manual approaches to ensure that the project meets functional requirements and maintains high reliability. The following types of testing will be

performed:

- **Unit Testing:** We will implement unit tests for critical components, including geofence logic and location-based features. Pytest will be the primary testing framework. Unit testing will begin as soon as core modules are developed and will continue throughout the project.
- **Integration Testing:** Integration testing will verify interactions between different system components, such as API calls to The Things Stack and web service calls to the Google Maps API. These tests will be mocked where appropriate to ensure reliability and repeatability.
- **System Testing:** End-to-end system testing will be conducted once a minimum viable product is available. This will include route pings, location checking, and verifying that hosted meals can be created, displayed, and joined.
- **Acceptance Testing:** Acceptance testing will be performed in collaboration with Dr. Damiano Zanutto. This testing will validate that the system meets the agreed project requirements and is ready for delivery.
- **Regression Testing:** Regression tests will be rerun whenever bug fixes or new features are introduced to ensure that existing functionality remains stable.
- **Testing for Quality Attributes:** While performance and security are not the primary focus, geolocation accuracy and reliability will be emphasized. Specific attention will be given to ensuring that geofences trigger consistently and API interactions are fault-tolerant.
- **Continuous Integration (CI):** CircleCI will be used to automate testing during development. All code changes will trigger automated test runs to catch issues early.

Our stopping criteria for testing is defined as achieving at least 80% unit test coverage across critical modules, successful integration of geofencing and API functionality, and passing acceptance testing with project stakeholders.

3.8 Risks

Risks that we must mitigate throughout the development of our product include various issues on all scales of development. With the development board, we want to ensure that the battery life is competitive with existing tracking units, such as AirTag. We also would like to ensure that the tracking unit is able to provide precise location data for the user to accurately locate the unit.

3.9 Assumptions (required)

For the development of our program, it is assumed that:

- Publicly accessible LoRaWAN gateways populate our region, allowing for fluid connection between our board to LoRaWAN network servers.
- End users have mobile devices or computers that are able to access our website, providing them access to their tracking units.

3.10 DISTRIBUTION LIST

This document will be shared with the members of the FindIt team, as well as Dr. David Darian Muresan and Dr. Damiano Zanotto throughout our development processes.

3.11 IRB Protocol

Our project does not require IRB protocol.

3.12 Required Resource and Budget

3.12.1 Hardware Resources

For our tracking unit, the only development expense will include the Wio Tracker 1110 development board, costing \$30.

3.12.2 Software Resources

Depending on the size of our infrastructure, deployment on AWS as well as the storage of historical data on Supabase will require subscriptions in order to maintain the usage of our software.

Chapter 4

Requirements

– Ryan Davis, Cory Vitanza, and Roy Tas

4.1 Stakeholders

- Wearable Robotics Systems Lab
- Assisted Living Homes
- Patrons with autism or dementia

4.1.1 Customers

- Families with autistic members
- Caregivers for dementia patients
- Assisted living homes

4.1.2 Sponsors

- Wearable Robotic Systems Lab

- Stevens Institute of Technology

4.1.3 Competitors

Device	Price & Subscription	Connection Type	Target Users	Key Features
Angel Sense	\$220 + \$45/month	Cellular	Autism / Children	Guardians, alerts & geofences, live safe ride transit tracking
Jiobit Smart Tag	\$140 + \$10/month	Cellular	Elderly, children, autism	Location tracking
Medical Guardian	\$150 + \$40/month	Cellular	Elderly	SOS button directly connecting to emergency operators, fall detection

Table 4.1: Comparison of Elderly and Autism Tracking Devices

4.2 Key Concepts

- [LoRaWAN \(Long Range Wide Area Network\)](#) enables long-range, low-power communication.
- [GPS \(Global Positioning System\)](#) provides location coordinates for tracker position.
- [Wio Tracker 1110](#) serves as the hardware platform.
- [Supabase](#) stores GPS data and user configurations.
- [RSSI \(Received Signal Strength Indicator\)](#) measures LoRaWAN signal strength.
- [Device Uplink](#) sends GPS data to the backend.
- [Device Downlink](#) updates device configuration remotely.
- [Gateway](#) receives device transmissions.

- [The Things Stack \(TTS\)](#) (*The Things Stack*) receives payloads and forwards them to Supabase.

4.3 User Requirements

Table 4.2: User Requirements Table

Requirement	Priority	Use Case(s)
FR1 (reqfGeofence): The system shall support geofencing with alerts when a tracker exits predefined boundaries.	MustHave	UC1
FR2 (reqfRealtimeUpdate): The system shall provide real-time location updates via the caregiver mobile app.	MustHave	UC1
FR3 (reqfMultiDevice): The system shall support simultaneous monitoring of multiple trackers.	MustHave	UC2
FR4 (reqfConfigNotifications): The system shall allow configurable notifications for multiple users or devices.	ShouldHave	UC2
FR5 (reqfBatteryMonitor): The device shall monitor remaining battery capacity and send low-power alerts.	MustHave	UC3
FR6 (reqfBatteryLife): The system shall maintain battery life comparable to consumer devices such as Apple AirTag.	CouldHave	UC3
FR7 (reqfEmergencyButton): The system shall include a button that triggers an emergency alert notification to caregivers.	ShouldHave	UC5
FR8 (reqfPatternDetection): The system shall use a machine learning algorithm to detect abnormal movement patterns.	ShouldHave	UC6

Table 4.2: User Requirements Table

Requirement	Priority	Use Case(s)
CR1 (reqcNoSubscription): The system shall operate without subscription fees by using community LoRaWAN gateways.	MustHave	UC4
QR1 (reqqScalability): The system shall scale to support multiple concurrent devices and users (about 50 devices per group).	ShouldHave	UC4
QR2 (reqqModelPerformance): The ML model shall maintain accuracy in detecting anomalies without degrading battery life.	CouldHave	UC6

4.4 System (Constraints) Requirements

Table 4.3: System (Constraints) Requirements

Requirement	Priority	Type
SR1 (reqsConnectivity): The system shall connect to at least one LoRaWAN gateway within a 2 km radius under normal conditions.	MustHave	Functional
SR2 (reqsTransmission): The tracker shall send GPS coordinates via uplink every configurable interval between 1–10 minutes.	MustHave	Functional
SR3 (reqsAPI): Supabase shall provide a secure RESTful API for data storage and retrieval.	MustHave	Functional
SR4 (reqsBattery): The tracker shall report battery status with each payload.	ShouldHave	Functional
SR5 (reqsEmergency): The emergency button shall send an uplink event within 5 seconds of activation.	MustHave	Functional
SR6 (reqsConfig): Downlink commands shall allow caregivers to modify device settings remotely.	ShouldHave	Functional
SR7 (reqsSecurity): All communications between Supabase and TTS shall use encrypted HTTPS endpoints.	MustHave	Security

Table 4.3: System (Constraints) Requirements

Requirement	Priority	Type
SR8 (reqsRetention): Supabase shall store location history data for at least 30 days.	ShouldHave	Functional
SR9 (reqsOffline): If no gateway is available, the tracker shall cache up to 10 unsent messages until reconnection.	ShouldHave	Reliability

4.5 Non-functional (Quality) Requirements

Table 4.4: Non-functional (Quality) Requirements

Requirement	Priority	Category
NFR1 (reqnPerformance): Uplink latency between device transmission and database update shall not exceed 15 seconds.	MustHave	Performance
NFR2 (reqnUsability): The caregiver app shall display alerts within two screen interactions.	ShouldHave	Usability
NFR3 (reqnMaintainability): The system shall be modular, allowing firmware or API updates independently.	ShouldHave	Maintainability
NFR4 (reqnSecurity): All stored user data shall follow AES-256 encryption standards.	MustHave	Security
NFR5 (reqnPower): The device shall operate for at least 5 days on a single charge at 15-minute intervals.	ShouldHave	Efficiency

4.6 Domain (Business) Requirements

Table 4.5: Domain (Business) Requirements

Requirement	Priority	Type
BR1 (reqbCompliance): The system shall comply with HIPAA regulations if identifiable patient data is stored.	MustHave	Legal
BR2 (reqbOpenInfra): The project shall rely on public Lo-RaWAN gateways instead of proprietary cellular networks.	ShouldHave	Business
BR3 (reqbOwnership): All user data shall remain the property of the caregiver or family account holder.	MustHave	Policy
BR4 (reqbMaintenance): Hardware upkeep shall be the responsibility of the owning institution or family.	ShouldHave	Operational

4.7 Requirements to Use Case Mapping

Table 4.6: Requirements, Use Case, and User Story Traceability Matrix

Req ID	Use Case(s)	User Story(ies)	Description
reqfGeofence	UC1	US1	Geofence alert notifications.
reqfMultiDevice	UC2	US2	Multi-resident caregiver monitoring.
reqfBatteryMonitor	UC3	US3	Battery monitoring alerts.
reqfEmergencyButton	UC5	US4	Emergency button response.
reqcNoSubscription	UC4	US5	Cost-efficient infrastructure.
reqqModelPerformance	UC6	US6	ML anomaly detection accuracy.

Chapter 5

User Stories

– Ryan Davis, Cory Vitanza, and Roy Tas

5.1 Overview

This chapter presents key user stories that describe the real-world needs and motivations of potential users of the tracking system. These stories help guide the system’s design and ensure that functional and non-functional requirements align with user expectations.

5.2 User Stories

5.2.1 Parent of a Child with Autism

- **US1: As a parent of a child with autism,** I want a device that automatically alerts me when my child leaves a safe zone so that I can respond quickly without wasting time searching.

5.2.2 Caregiver in an Assisted Living Facility

- **US2: As a caregiver responsible for multiple residents,** I want to track multiple devices simultaneously and receive alerts when a resident leaves a designated safe area so that I can act promptly to

ensure their safety.

5.2.3 Family Member Monitoring Battery Life

- **US3: As a family member monitoring a loved one,** I want to receive notifications when the tracker's battery is running low so that I can prevent the device from powering off unexpectedly during an emergency.

5.2.4 User Triggering an Emergency Alert

- **US4: As a user wearing the device,** I want to press a button in an emergency to send an alert to my caregiver or family members so that I can get help quickly.

5.2.5 System Administrator Managing Devices

- **US5: As a system administrator,** I want to manage LoRaWAN gateways and device registrations efficiently so that the system remains scalable and reliable without subscription costs.

5.2.6 AI Monitoring and Pattern Detection

- **US6: As a system backend operator,** I want the system to automatically detect unusual movement patterns so that potential emergencies can be identified proactively.

5.3 User Story to Use Case Mapping

Table 5.1: User Story to Use Case Mapping

User Story ID	Description	Mapped Use Case(s)
US1	Parent receives alerts when a child leaves a safe zone.	UC1

US2	Caregiver monitors multiple residents and receives alerts.	UC2
US3	Family member receives low-battery notifications.	UC3
US4	User triggers an emergency alert to notify caregiver.	UC5
US5	System administrator manages deployments and gateways.	UC4
US6	Backend detects unusual behavior through ML models.	UC6

5.4 Traceability Discussion

All user stories directly support at least one system use case to ensure end-to-end traceability. Each story reflects the needs of a real stakeholder—caregivers, users, family members, or system administrators—and guides the functional and non-functional requirements of the FindIt asset tracking system. By maintaining explicit links (**US1–US6**) between stories and use cases, this document ensures future system iterations remain aligned with user needs.

Chapter 6

Use Cases

– Ryan Davis, Roy Tas, Cory Vitanza

This chapter presents the use case diagrams that satisfy the requirements of the FindIt asset tracking system. Each use case represents a key user interaction scenario that drives the system requirements. All use cases are mapped to at least one requirement to ensure complete traceability.

6.1 Table of Use Cases

Table 6.1: Use Cases Table

Use Case	Requirements	Name and Description
UC1	reqfGeofence, reqfRealtimeUpdate	Geofence Alert – The system detects when the tracker exits or enters a predefined boundary and sends an alert to the caregiver’s phone.
UC2	reqfMultiDevice, reqfConfigNotifications	Multi-Resident Monitoring – A caregiver monitors multiple trackers simultaneously and receives notifications when any resident leaves a safe area.

Use Case	Requirements	Name and Description
UC3	reqfBatteryMonitor , reqfBatteryLife	Battery Status Alert – The tracker measures its remaining battery and sends a notification when power is low.
UC4	reqcNoSubscription , reqqScalability	Cost-Efficient Deployment – The system operates without subscription costs, relying on LoRaWAN gateways, and supports scaling across many devices.
UC5	reqfEmergencyButton	Emergency Alert – A user can trigger an alert in case of an emergency, notifying caregivers or family members for immediate response.
UC6	reqfPatternDetection , reqqModelPerformance	Machine Learning Pattern Detection – The system analyzes historical tracker data to identify unusual movement or location patterns.

6.2 Use Case Diagram

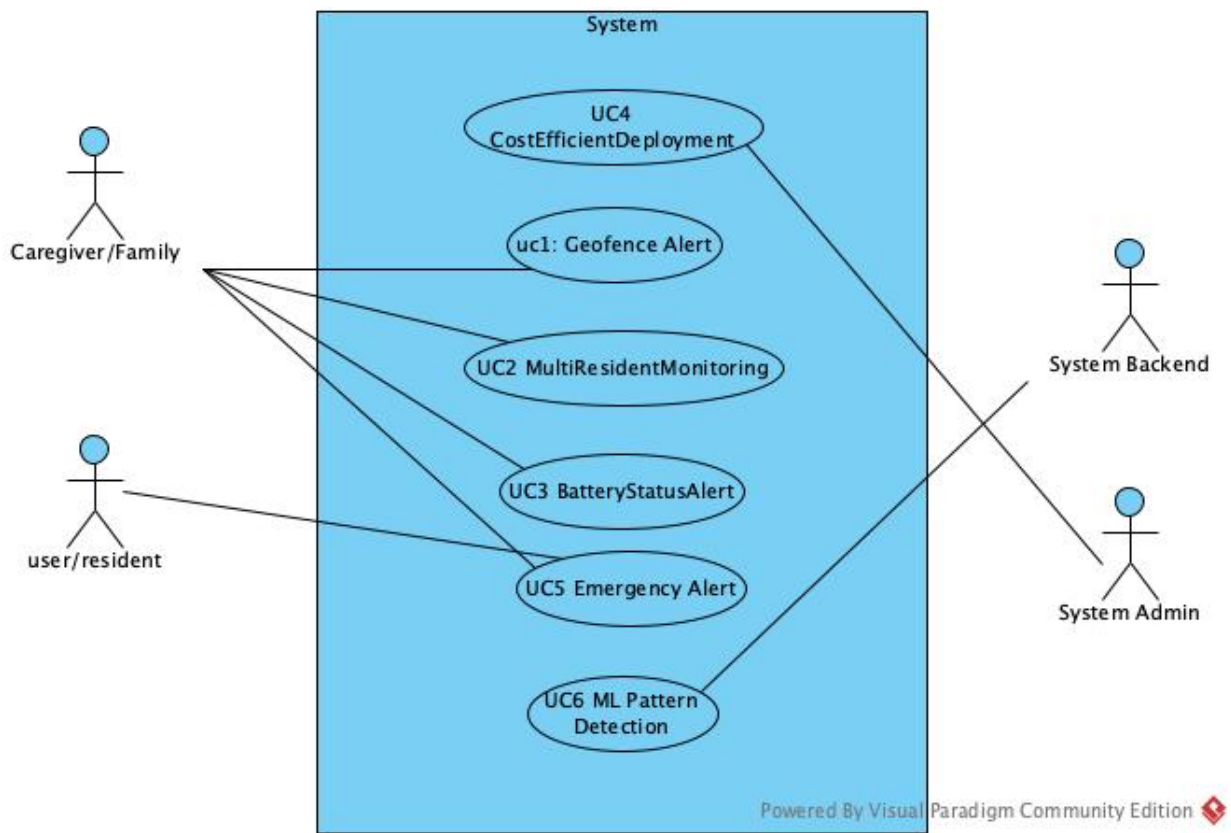


Figure 6.1: FindIt Use case diagram.

6.3 Use Case Descriptions

Table 6.2: UC1. Geofence Alert

UC1 Geofence Alert – Detects when the tracker exits a predefined area and notifies the caregiver.
Requirements: reqfGeofence , reqfRealtimeUpdate
Primary actors: Caregiver, Tracker Device
Preconditions: Tracker and caregiver app are connected to the LoRaWAN network.
Main flow: 1. The caregiver defines a geofence area. 2. The tracker transmits periodic GPS data. 3. The system detects an exit from the geofence. 4. A notification is sent to the caregiver.
Postconditions: Caregiver receives alert in near-real time.

Table 6.3: UC2. Multi-Resident Monitoring

UC2 Multi-Resident Monitoring – Allows caregivers to view and track multiple residents simultaneously.
Requirements: reqfMultiDevice , reqfConfigNotifications
Primary actors: Caregiver
Preconditions: Multiple FindIt trackers are registered to one caregiver account.
Main flow: 1. The caregiver logs into the FindIt dashboard. 2. Multiple devices report position updates. 3. The caregiver views the dashboard for all trackers.
Postconditions: The caregiver can identify any tracker that leaves its assigned safe zone.

Table 6.4: UC3. Battery Status Alert

UC3 Battery Status Alert – Sends low battery notifications to caregivers.
Requirements: reqfBatteryMonitor , reqfBatteryLife
Primary actors: Tracker Device, Caregiver
Main flow: 1. Tracker measures remaining battery capacity. 2. Device sends low battery alert. 3. Caregiver recharges or replaces device.
Postconditions: Battery health is maintained.

Table 6.5: UC4. Cost-Efficient Deployment

UC4 Cost-Efficient Deployment – Uses community LoRaWAN networks and low-cost back-end services.
Requirements: reqcNoSubscription , reqqScalability
Primary actors: System Administrator, Caregiver
Main flow: 1. Tracker connects via community gateway. 2. Data stored in Supabase. 3. System scales to additional users.
Postconditions: Low operational cost and scalability.

Table 6.6: UC5. Emergency Alert

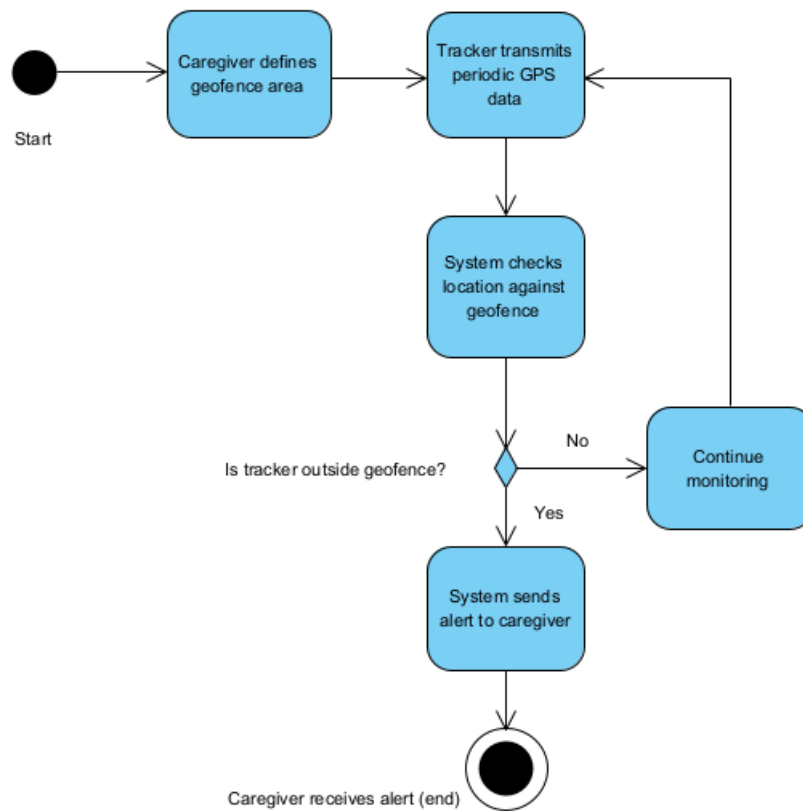
UC5 Emergency Alert – Allows users to trigger an emergency notification.
Requirements: reqfEmergencyButton
Primary actors: User, Caregiver
Main flow: 1. User presses emergency button. 2. Tracker sends emergency alert. 3. Caregiver is notified.
Postconditions: Emergency is communicated promptly.

Table 6.7: UC6. Machine Learning Pattern Detection

UC6 Machine Learning Pattern Detection – Identifies abnormal movement patterns using historical data.
Requirements: reqfPatternDetection , reqqModelPerformance
Primary actors: Caregiver, Backend System
Main flow: 1. Tracker uploads historical data. 2. Backend analyzes movement trends. 3. Deviations are detected. 4. Caregiver is alerted.
Postconditions: Early detection of abnormal behavior.

6.4 Activity Diagrams

This section presents activity diagrams corresponding to each primary use case. Each diagram illustrates the runtime process flow that realizes the behavior described in its associated use case.

Figure 6.2: Activity Diagram for **UC1** (Geofence Alert)

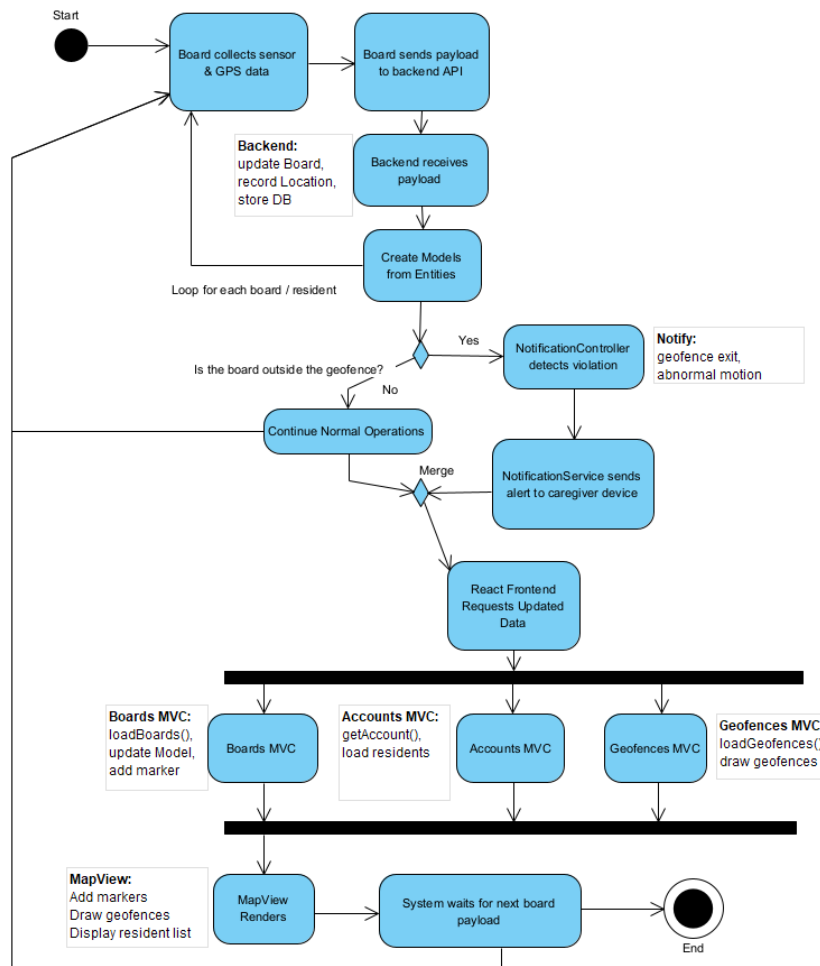
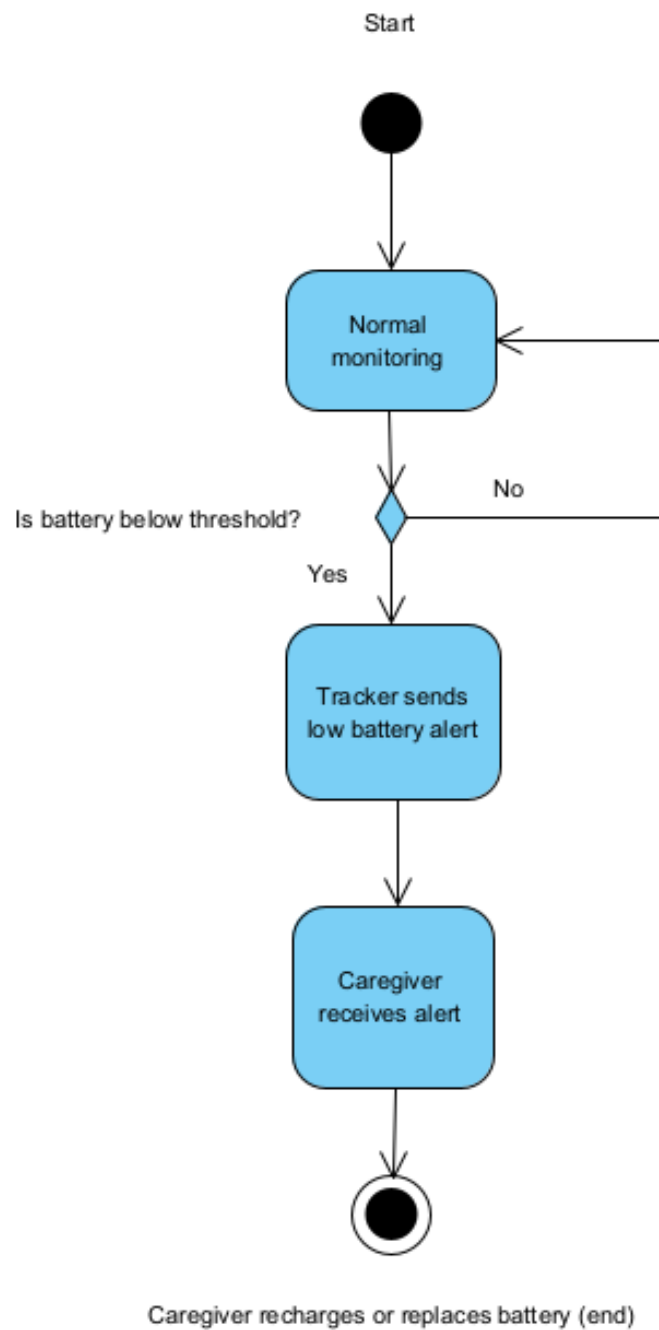
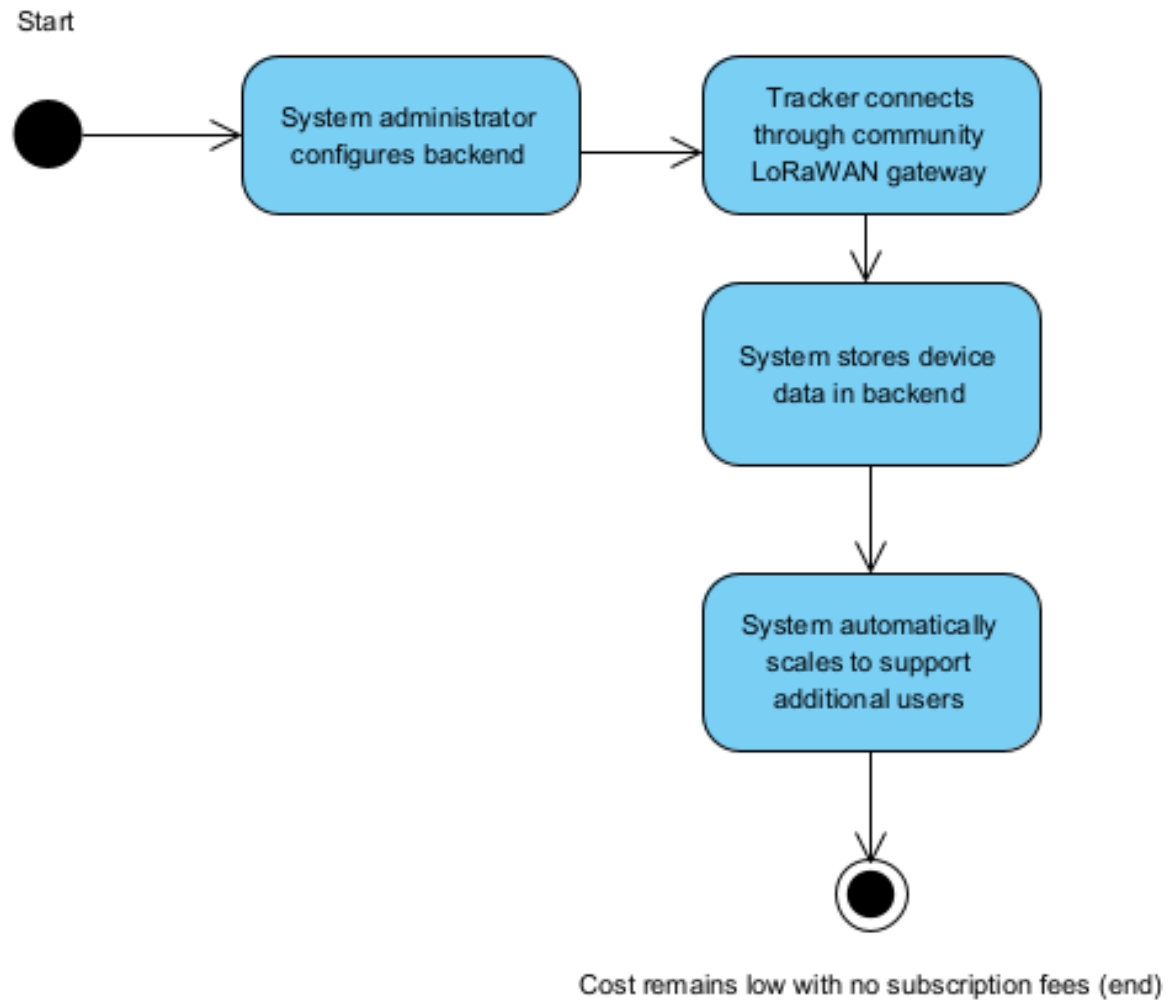
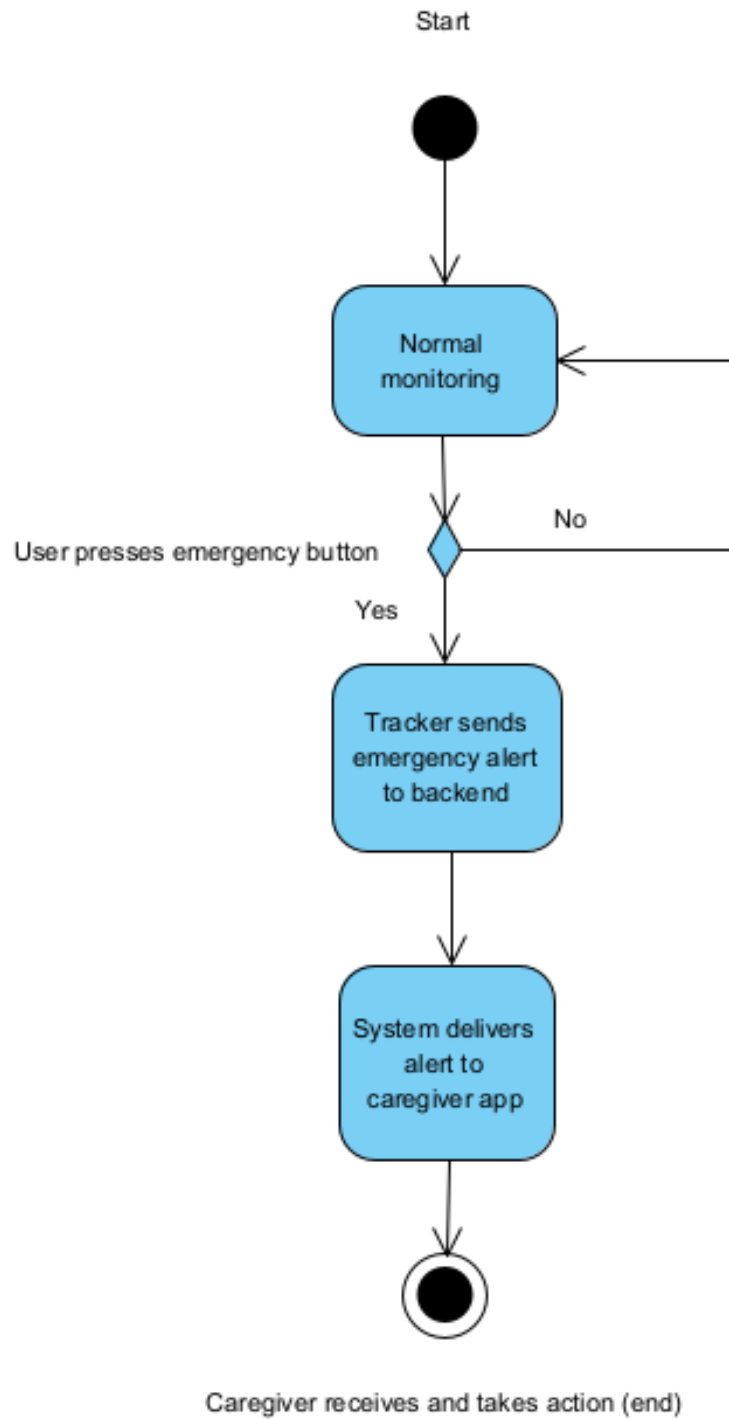
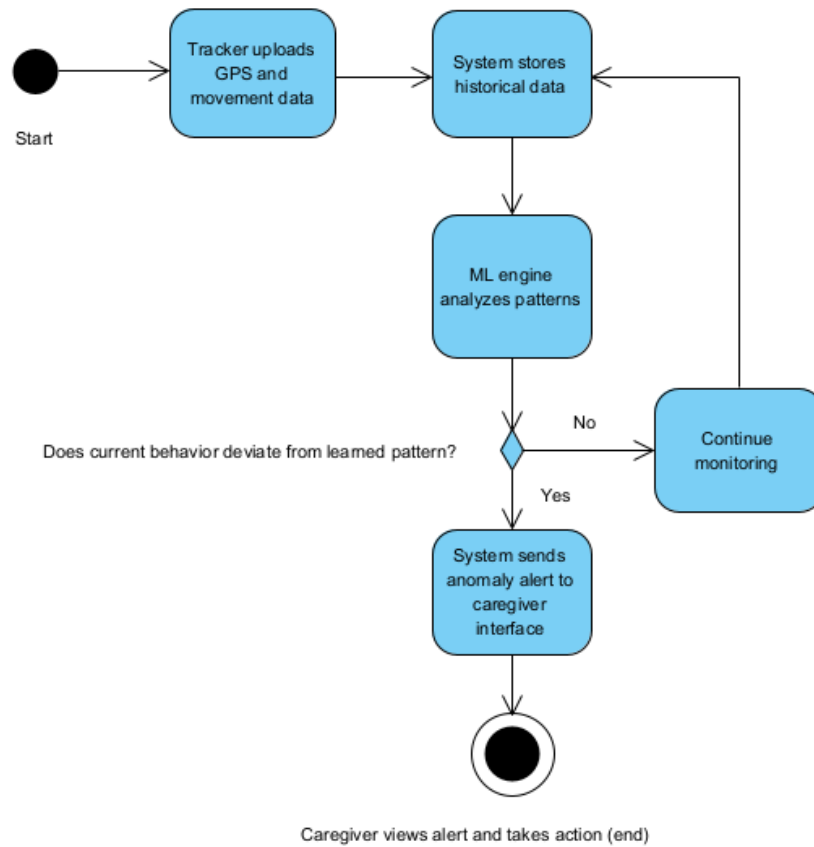


Figure 6.3: Activity Diagram for UC2 (Multi-Resident Monitoring)

Figure 6.4: Activity Diagram for **UC3** (Battery Status Alert)

Figure 6.5: Activity Diagram for **UC4** (Cost-Efficient Deployment)

Figure 6.6: Activity Diagram for **UC5** (Emergency Alert)

Figure 6.7: Activity Diagram for **UC6** (Pattern Detection)

Chapter 7

User Interface Design

– Ryan Davis, Roy Tas, and Cory Vitanza

7.1 User Persona

Maria Davis

- **Age:** 42 years old
- **Family Status:** Full-time caregiver for her 80-year-old mother (early dementia)
- **Location:** Brooklyn, NY
- **Occupation:** Part-time dental assistant
- **Background:** Moved her mother in last year after wandering incidents
- **Core Challenge:** Balancing work, kids, and caregiving; high morning anxiety
- **Required Solution Features:**
 - Simple and dependable device
 - Immediate geo-fence notifications
 - Reduced need for continuous supervision

John Smith

- **Age:** 10 years old
- **Condition:** Diagnosed with autism
- **Location:** Paramus, NJ
- **Wandering Triggers:** Stressful or crowded environments (malls, airports)
- **Personality:** Smart, curious, active; enjoys technology
- **Device Preferences:**
 - Prefers small wearable devices
 - Dislikes bulky devices
- **Required Solution Features:**
 - Real-time location updates for parents
 - Quick distance + geo-fence alerts
 - Simple UI for school staff

Brian Cleary

- **Age:** 67 years old
- **Status:** Retired mechanic living alone
- **Location:** Weehawken, NJ
- **Health Concerns:** Dizziness; fall risk; forgets to charge devices
- **Routine:** Morning walks by the Hudson River
- **Device Preferences:**

- Prefers extremely simple or automatic devices
- Avoids smartphones
- **Required Solution Features:**
 - Long battery life
 - Automatic fall detection
 - Passive well-being updates for family

7.2 User Interface Design

Figma Paper Prototype: <https://www.figma.com/make/jALgOM7hNgKhoa1sMRlw5L/User-Interface-for-Tracking-node-id=0-4>

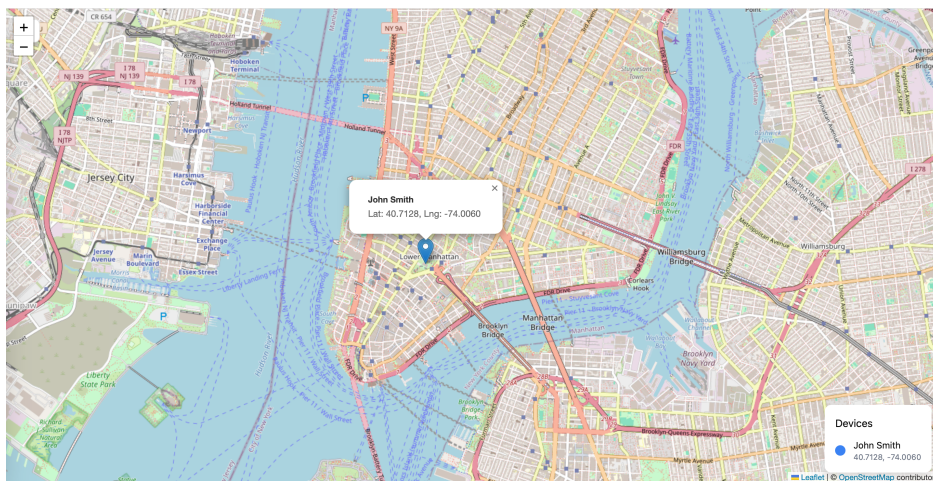


Figure 7.1: PaperPrototype: Figma Paper Prototype for FindIt UI.

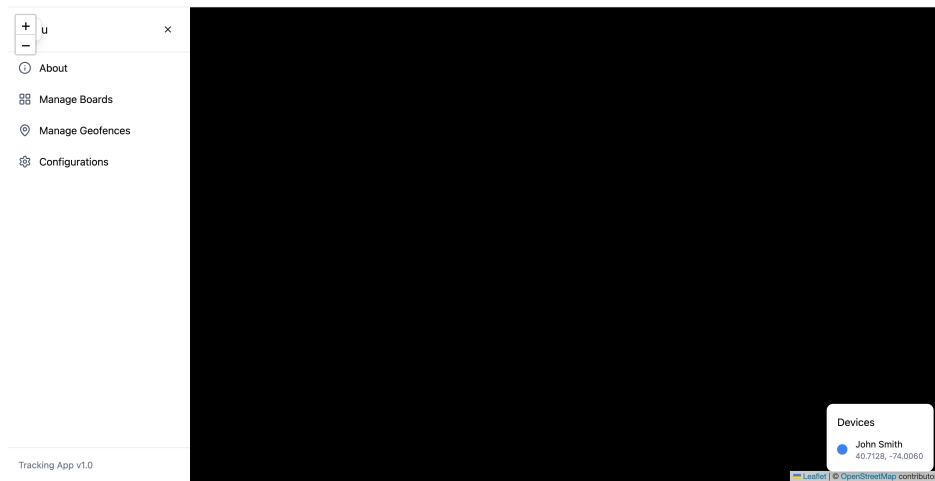


Figure 7.2: PaperPrototype: Figma Paper Prototype of Side Menu.

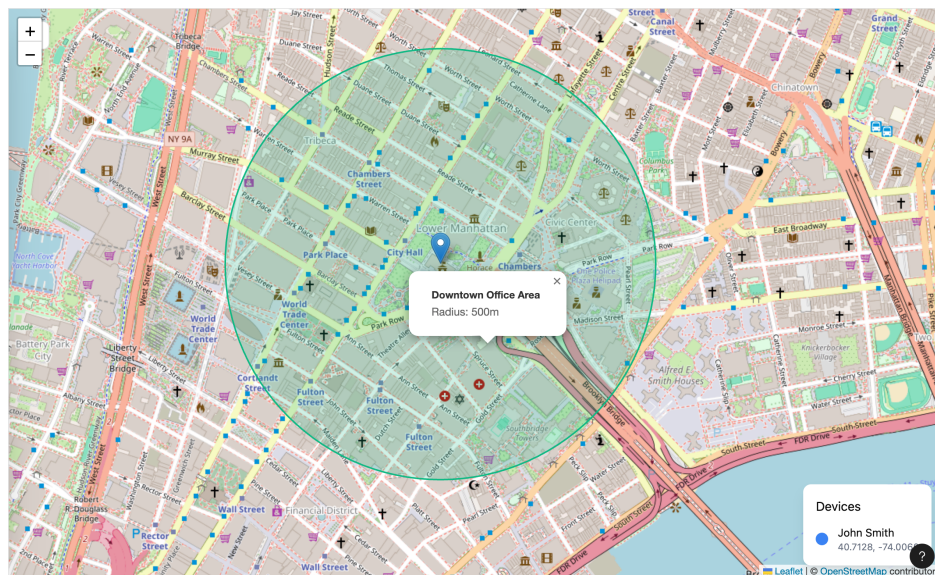


Figure 7.3: PaperPrototype: Figma Paper Prototype of Geofence Implementation.

Chapter 8

Logical View

– Ryan Davis, Roy Tas, and Cory Vitanza

Class diagrams present the structure of our Design, highlighting the firmware on our Wio Tracker 1110, and how that data is processed in our React software.

Class Diagram Description

- **Overview**

- The class diagram is organized into four primary packages: **Entities**, **Models**, **Controllers**, and **Views**.
- The system follows the Model–View–Controller (MVC) architectural pattern as implemented in the React frontend.
- Each package has a clearly defined responsibility, and interactions between packages are explicitly controlled to maintain separation of concerns.

- **Entities Package**

- The Entities package contains domain objects that directly represent rows in the Supabase database.

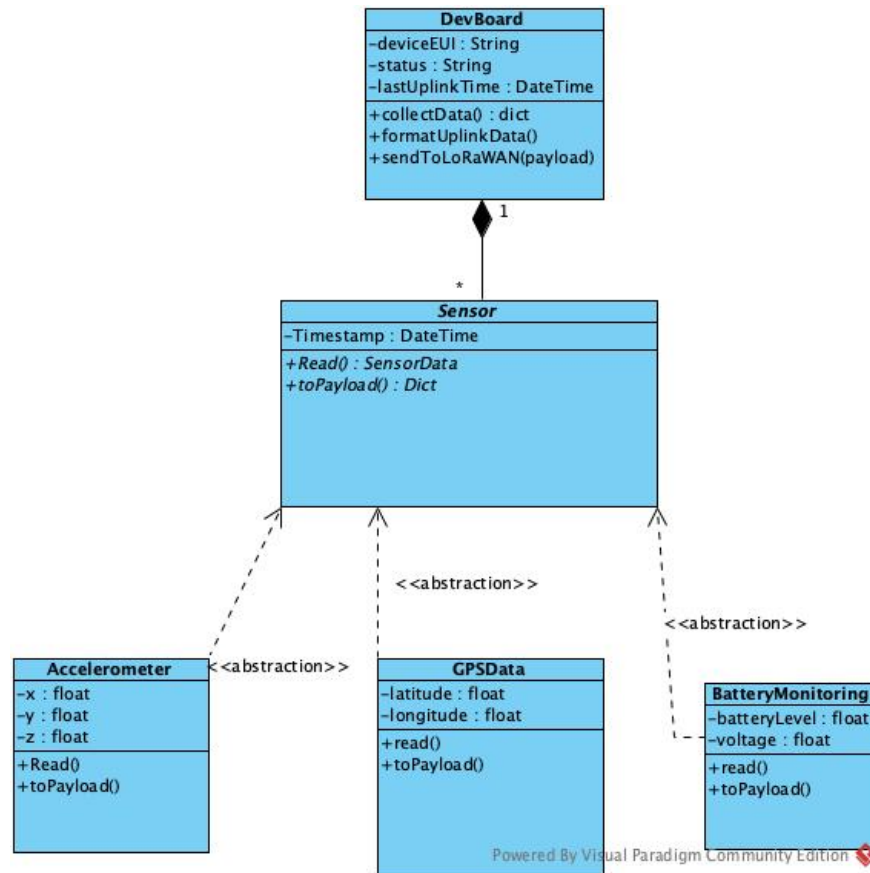


Figure 8.1: FirmwareClasses: Class diagram for the Wio Tracker 1110 Development Board's Firmware

- Entities include FIUBoardEntity, FIUGeofenceEntity, FIUAccountEntity, and FIULocationRecordEntity.
- All entities implement the FIUEntity interface, which standardizes the id attribute as a UUID.
- Although UUIDs are serialized as strings in the React application, they are modeled as UUIDs at the domain and database level.
- Entities contain no business logic and are used exclusively as raw data containers.

• Models Package

- Models encapsulate Entities through composition and extend the abstract FIUModel base class.
- Models are responsible for domain logic, validation, and data transformation required by the UI.

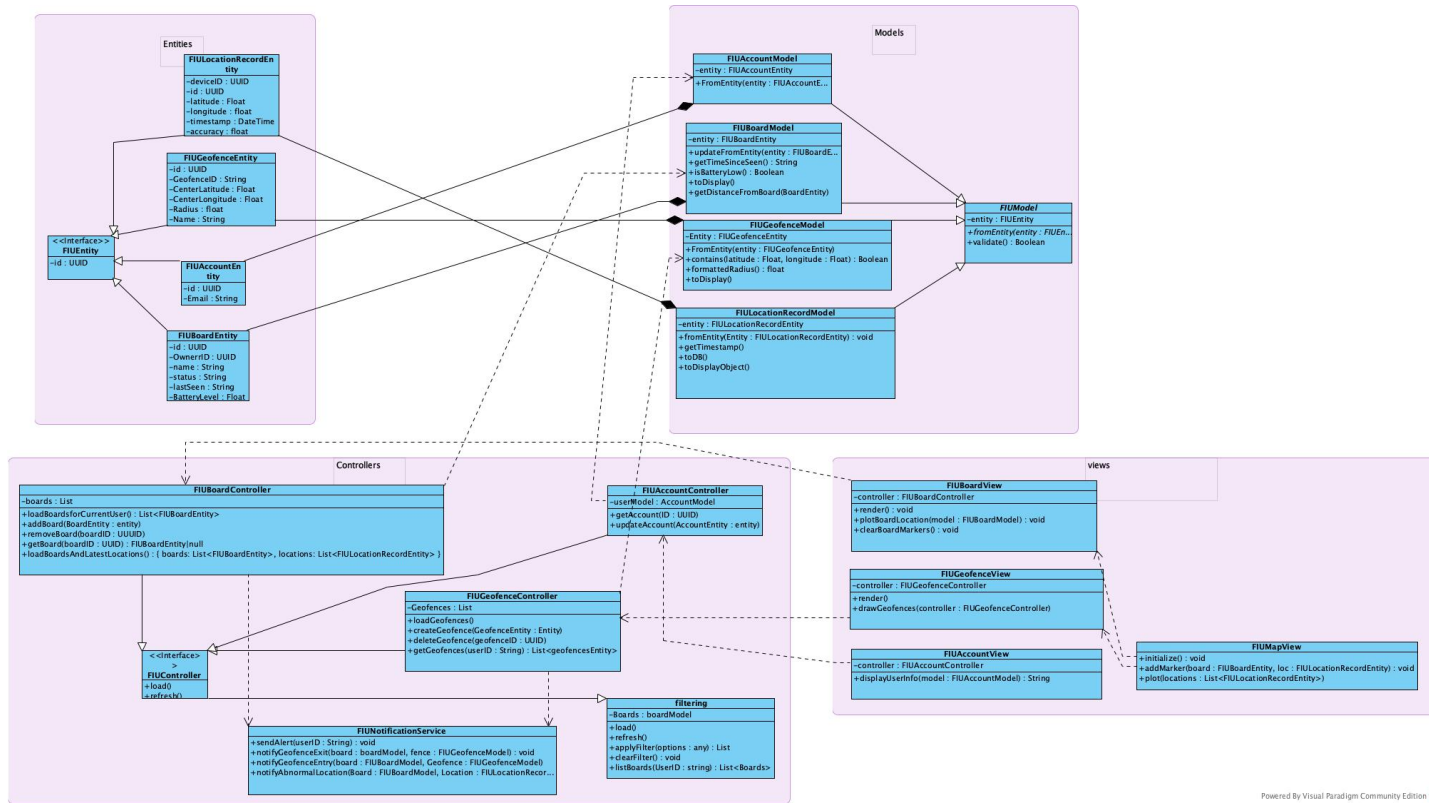


Figure 8.2: Model View Controller: Class Diagram for our React Interface Using the model view controller design pattern.

- Examples of model responsibilities include computing geofence containment, formatting timestamps, calculating distances, and evaluating battery thresholds.
- The Models package includes `FIUBoardModel`, `FIUGeofenceModel`, `FIUAccountModel`, and `FIULocationRecordModel`.

• Controllers Package

- Controllers coordinate application logic and act as the sole intermediaries between the Views and the backend data layer.
- Controllers load data from Supabase via repository functions, instantiate Entities, and construct Models for presentation.
- The `FIUBoardController` orchestrates the primary dashboard workflow by loading boards

owned by the authenticated user and retrieving the latest location records for each board.

- Ownership-based access is the primary mechanism for board and location retrieval; membership-based access is optional and not required for the current implementation.
- Controllers never render UI elements directly and do not access the database from Views.

- **Views Package**

- Views are responsible solely for rendering UI components and delegating actions to their associated Controllers.
- Views receive prepared Models from Controllers and never access Entities or repositories directly.
- The `FIUMapView` renders board locations using Leaflet by plotting markers derived from `FIUBoardEntity` and `FIULocationRecordEntity` data.
- Views include `FIUBoardView`, `FIUGeofenceView`, `FIUAccountView`, and the shared `FIUMapView`.

- **Key Relationships**

- **Inheritance:**

- * All Model classes extend the abstract `FIUModel`.
- * Controllers implement the `FIUController` interface.

- **Composition:**

- * Each Model owns exactly one Entity instance (e.g., `FIUBoardModel` composes a `FIUBoardEntity`).

- **Dependencies:**

- * Views depend on Controllers.
- * Controllers depend on Models and backend repository functions.
- * Views depend on the shared map component for spatial visualization.

- **Separation of Concerns:**

- * Views do not contain business or domain logic.
- * Models do not perform rendering or data access.
- * Controllers coordinate logic and data flow but do not render UI elements.

- **Summary**

- The diagram accurately reflects the implemented MVC architecture of the FindIt frontend.
- Data flows from Supabase into Entities, is transformed by Models, coordinated by Controllers, and rendered by Views.
- This structure ensures maintainability, clear access control boundaries, and a clean separation between data, logic, and presentation layers.

Chapter 9

Development View

– Ryan Davis, Roy Tas, and Cory Vitanza

The package diagram for the FindIt system highlights the different modules in our system. The Wio Tracker 1110 has firmware, uplinking the data it collected into the database. The database provides data to our ML model highlighting abnormalities in location data, and to the React frontend, where the data is processed and presented on the user interface.

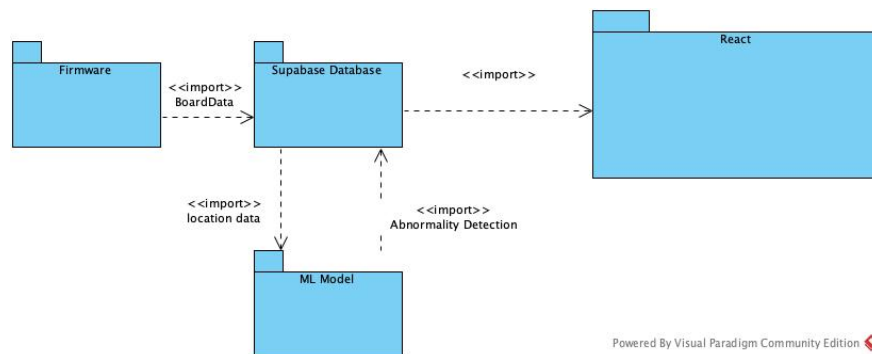


Figure 9.1: Package Diagram: Package diagram for FindIt deployment

Chapter 10

Process View

– Ryan Davis, Roy Tas, and Cory Vitanza

This sequence diagram shows the flows for our most prevalent use case, multi-resident monitoring. The activity starts with boards collecting gps data, and provides the end user with location data and notifications if a board leaves the geofence.

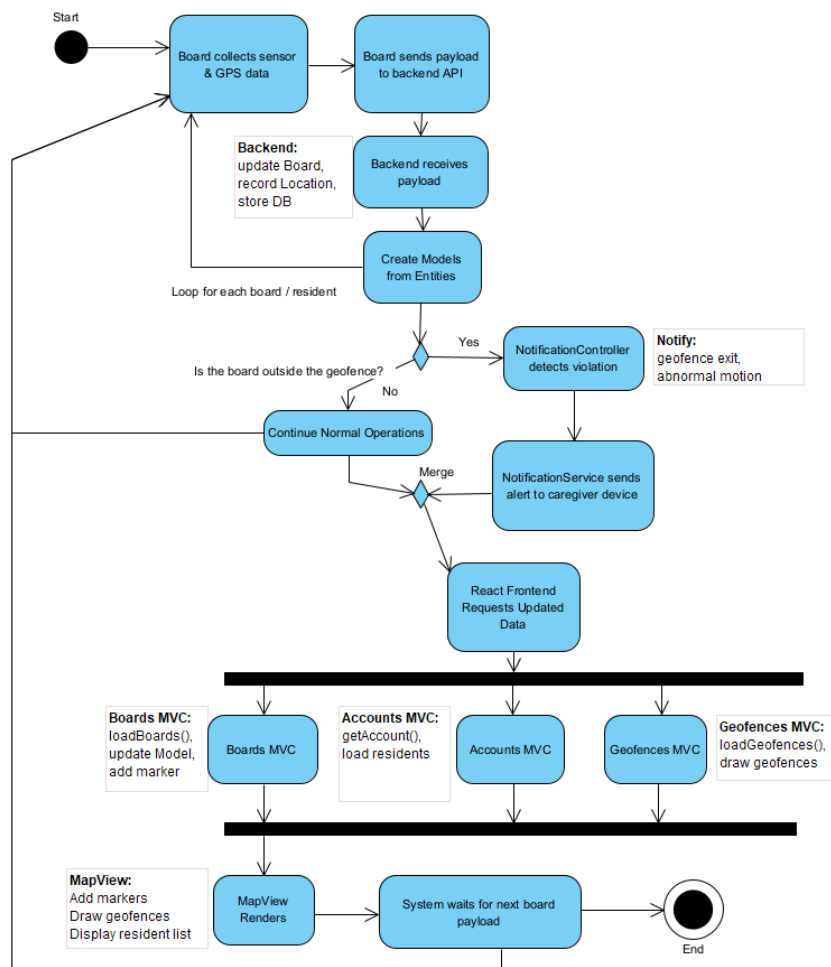


Figure 10.1: ucMultiResidentMonitoringActivity: Activity Diagram for Multi-Resident Monitoring (UC2)

Physical View

– *Ryan Davis, Roy Tas, and Cory Vitanza*

FindIt component diagram depicting the interfaces for the different components of the system, including the development board, database, backend infrastructure, and react components.

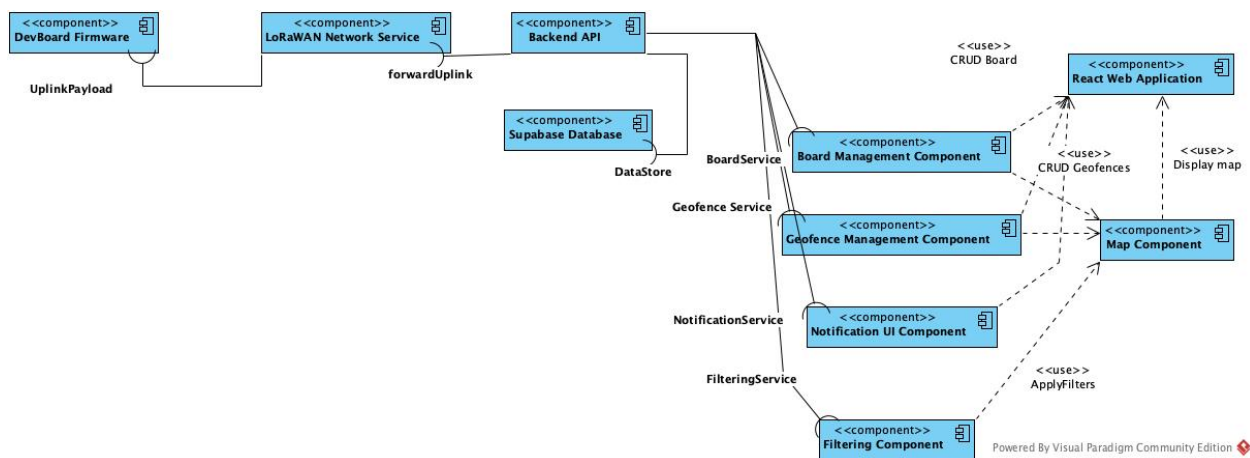


Figure 11.1: Component Diagram: Component diagram for FindIt deployment

Chapter 12

Glossary

Asset Tracking Monitoring the location of objects or people using GNSS (outdoor) and signal-based localization (indoor). [58](#)

Battery Life Optimization Design focus on minimizing energy usage through duty cycling, GNSS assist, and efficient LoRaWAN classes to target ≥ 1 year per device. [58](#)

CouldHave This defines the third highest priority requirement. The system could implement all of the tasks, requirements, or anything that is marked this way, but if resources are limited, it can be left out of the current and next version. Build in two versions from now. [21](#), [22](#), [58](#)

Device Downlink Network server transmits data to device. [19](#), [58](#)

Device Uplink Device transmit data to network server. [19](#), [58](#)

Emergency Communication User-initiated alerts (e.g., panic button or Morse-code sequence) transmitted via LoRaWAN for caregiver notification. [58](#)

FindIt System The end-to-end asset tracking solution developed by the team, consisting of the FindIt device, LoRaWAN gateway, and cloud-based data management dashboard. [58](#)

Firmware Embedded software on the tracker controlling sensing, power management, and LoRaWAN communication. [58](#)

Gateway A device that receives LoRaWAN packets from trackers and forwards them to the network server via the internet, bridging local RF to cloud. [19](#), [58](#)

Gateway Network Coverage Geographic area within which gateways can reliably receive tracker signals, determining communication reach and reliability. [58](#)

GPS (Global Positioning System) Satellite-based navigation providing precise outdoor positioning. Used by the tracker when sky visibility is sufficient. [19](#), [58](#)

Indoor Positioning Location estimation in GPS-denied spaces using RSSI triangulation, Wi-Fi fingerprinting, or Bluetooth beacons. [58](#)

LoRa (Long Range) A wireless modulation technique that enables long-range, low-power communication between IoT devices using sub-GHz frequency bands. [58](#)

LoRaWAN (Long Range Wide Area Network) A protocol built on top of LoRa that defines how end devices communicate with gateways and network servers for secure and scalable data transmission. [2](#), [19](#), [58](#)

LR1110 A Semtech transceiver that supports LoRa and GNSS assistance, enabling hybrid indoor/outdoor tracking and low-power operation. [58](#)

MustHave This defines the first highest priority requirement. All of the tasks, requirements, or anything that is marked this way are built in the current version. [21](#), [22](#), [23](#), [25](#), [26](#), [58](#)

One-Time Purchase Model Fixed device cost (e.g., \$50) with optional subscription tiers for extended history, private hosting, or premium features. [58](#)

RSSI (Received Signal Strength Indicator) Measured signal power used for distance/proximity inference and coarse indoor localization. [19](#), [58](#)

ShouldHave This defines the second highest priority requirement. The system should implement all of the tasks, requirements, or anything that is marked this way, but if resources are limited, it can be left out of the current version. Build in next version. [21](#), [22](#), [23](#), [24](#), [25](#), [26](#), [58](#)

Supabase The backend platform (PostgreSQL + REST/Realtime) used to store and query tracker data and power the FindIt API/dashboard. [19](#), [58](#)

The Things Stack (TTS) An open-source LoRaWAN network server for device registration, session management, and secure routing of uplink/downlink messages. [20](#), [58](#)

User Interface (UI) Mobile/web interface for viewing locations, receiving alerts, and managing settings. [58](#)

Wio Tracker 1110 The development board used for prototyping the FindIt device. It integrates Semtech's LR1110 transceiver, GNSS assistance, and LoRa communication interface. [19](#), [58](#)

WouldHave This defines the lowest priority requirement. The system would like to implement all of the tasks, requirements, or anything that is marked this way, but only if resources are available. It can be left out of all future versions. [58](#)

Appendix A

Weekly Reports

- Ryan Davis, Cory Vitanza, and Roy Tas

A.1 Week of 2026-01-30 (Week 1)

A.1.1 This Week's Progress

- Refactored
- Refactored the React frontend to utilize an MVC architecture. ([Commit Link](#))
- Initialized Digital Ocean Droplet for our Ubuntu Server.
- Updated Firmware to produce LoRaWAN uplinks and location and accelerometer data collection

A.1.2 Next Week's Tasks

- Configure MQTT script on Digital Ocean Droplet to retrieve board data from Sensecap dashboard to Supabase Database.
- Order a gps antenna for the Wio Tracker 1110 to collect location data.

- Create unit and regression tests for the system, covering data acquisition from the board, API calls from the database, and mapping the locations on the map.

A.1.3 Action Items

Task	Owner	Due Date	Status
Write script for SenseCAP MQTT-to-database pipeline on DigitalOcean Droplet	Ryan Davis	2026-02-03	Awaiting DigitalOcean Droplet
Order GPS antenna	Cory Vitanza	2/3/26	Pending.
Take board measurements for case construction	Roy Tas	2/3/26	Awaiting precise measurement tools and GPS antenna size.

A.1.4 Issues & Risks + Mitigation

Issue / Risk	Impact	Likelihood	Mitigation	Owner
GPS size constraints	Transportability of the prototype tracking unit	Slim	Build case around GPS	FindIt Team

A.1.5 Sprint Snapshot

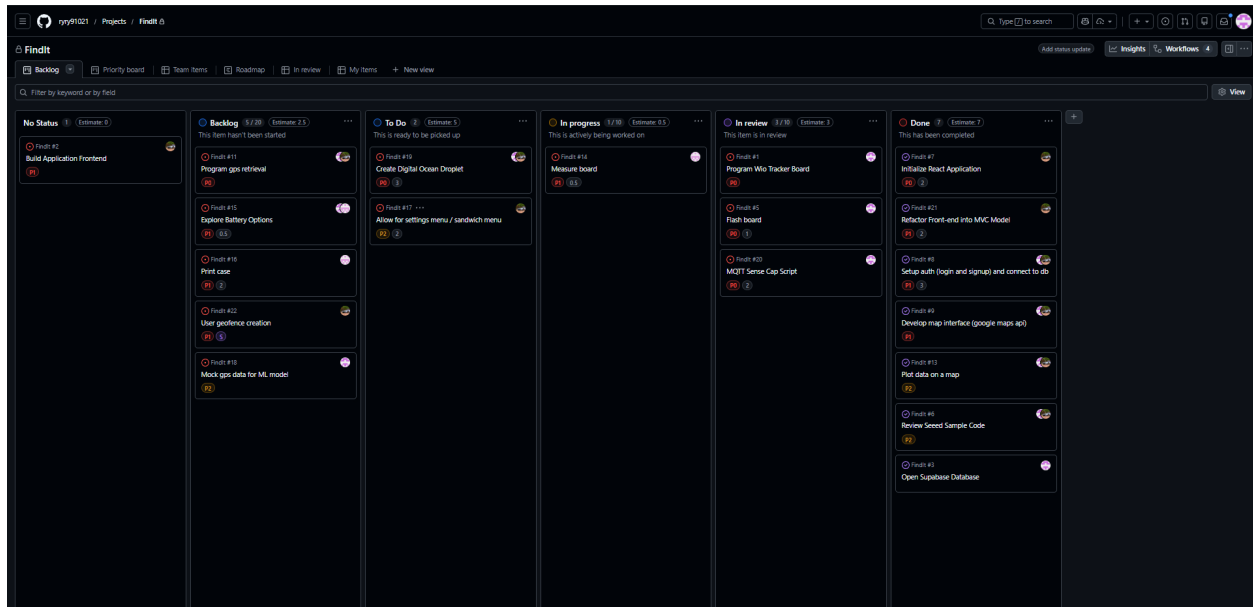


Figure A.1: Week report 1 Kanban Update

A.1.6 Diagram Work

- Updated [MVC Class Diagram](#) to account for development changes and Database schema updates.
- Updates include minor changes including namespace fixes, and method updates originating from refactoring and MVC construction.

A.2 Week Report 13 (12/11/2025)

A.2.1 Progress Made

- Connected board to SenseCraft UI for GPS testing.

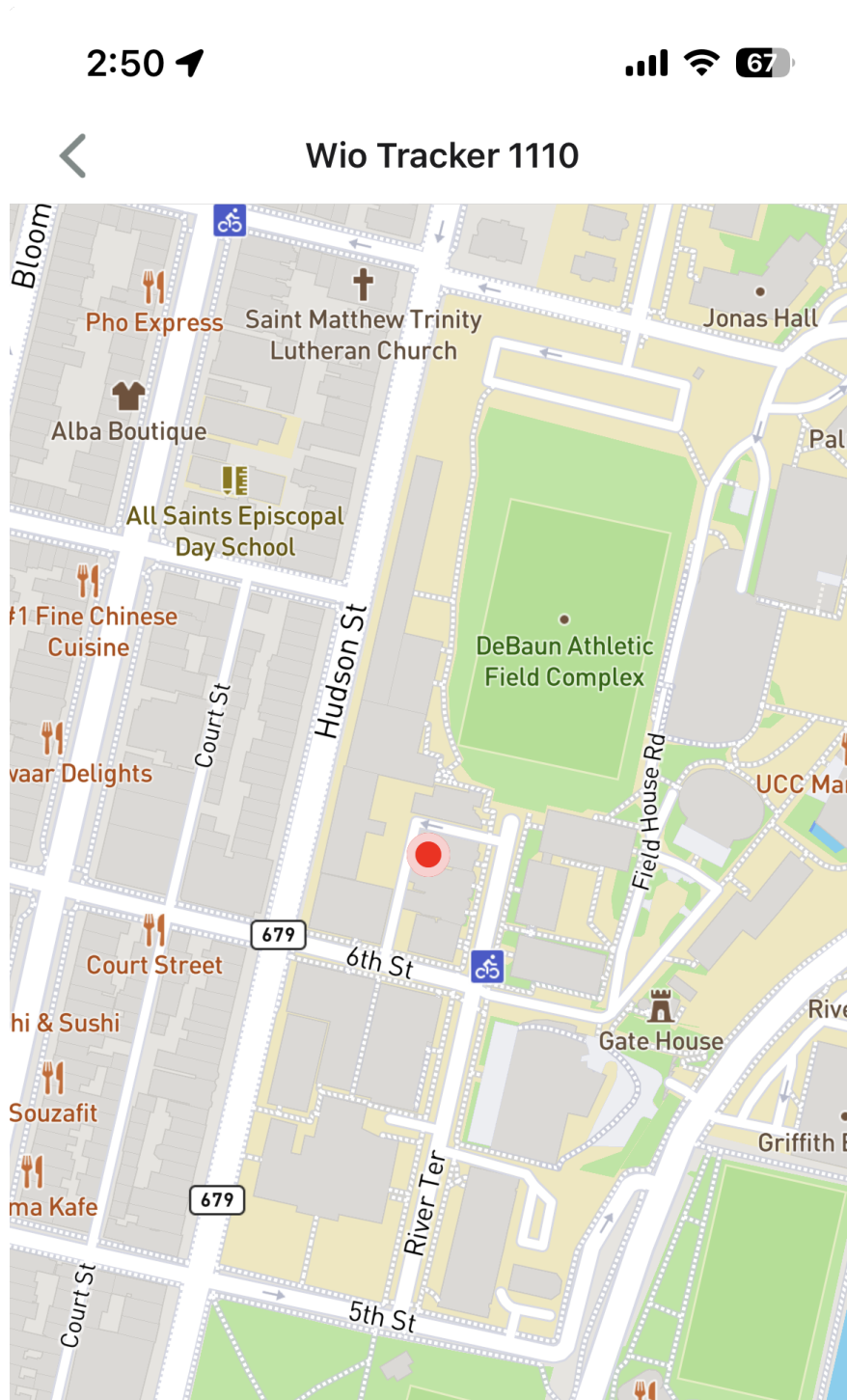


Figure A.2: SensecraftConnection: SenseCraft UI connection for GPS testing

The screenshot above shows the device's GPS output appearing correctly in SenseCraft.

- Developed authorization protocol with React Frontend and Supabase Database for user accounts.

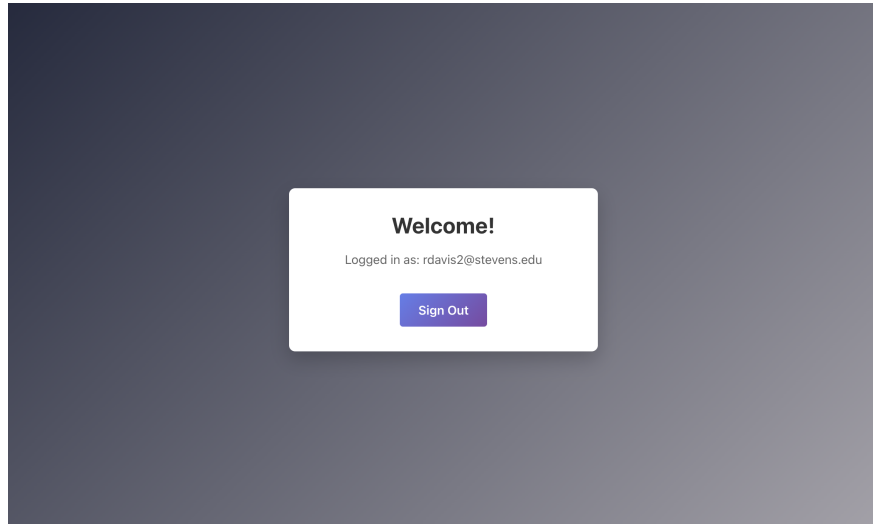


Figure A.3: DevServer: React development server showing authorization flow

The figure above would show the login system functioning in the dev server.

- Established database schema for board integrations.

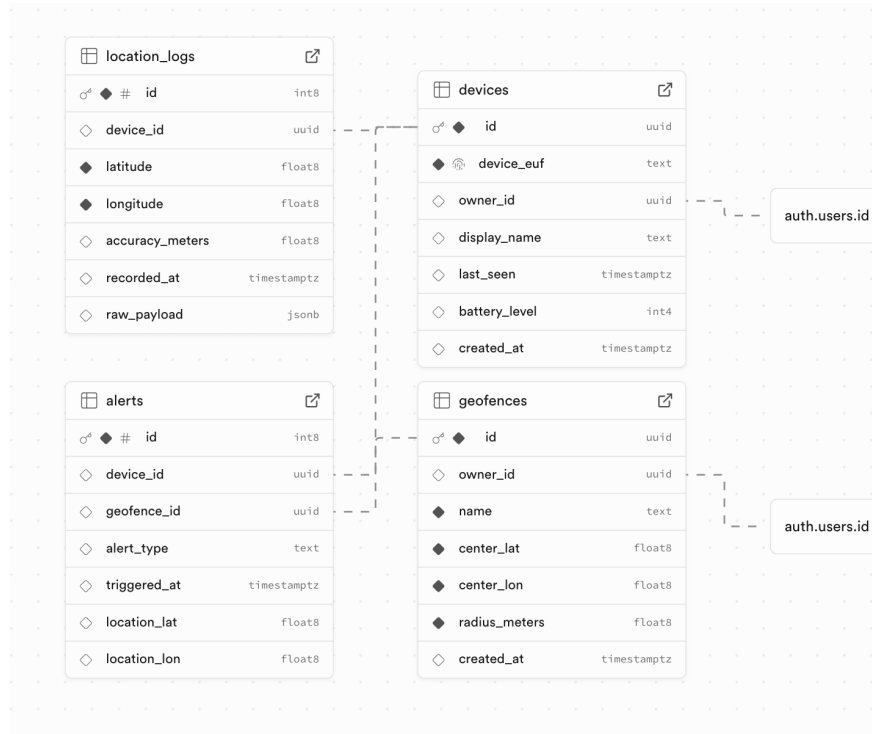


Figure A.4: SupabaseSchema: Supabase schema for board integration

The schema diagram above shows how each device is linked to a user via `auth.users.id`.

- Developed a case for the Wio Tracker 1110.

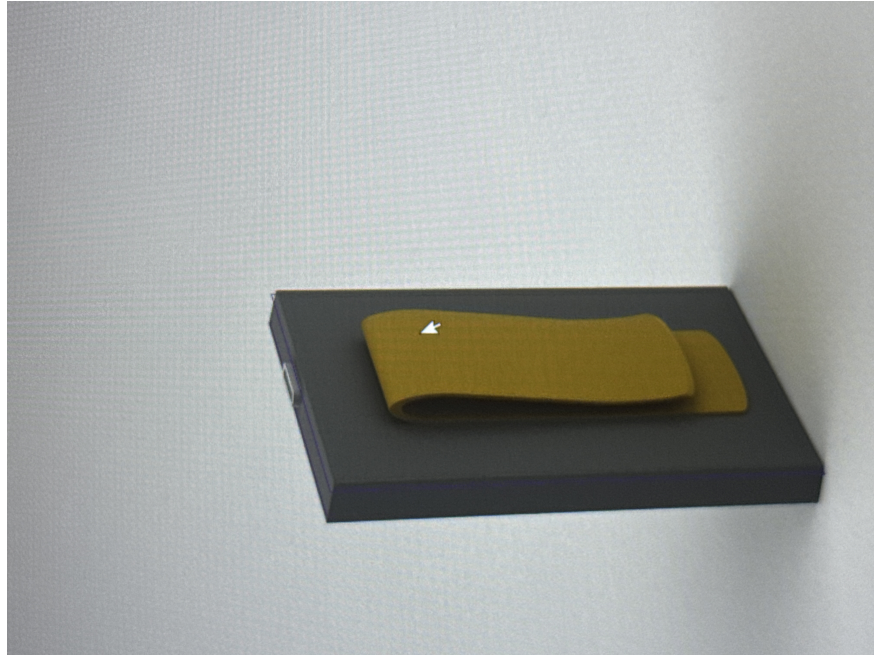


Figure A.5: CaseClip: Clip for the Wio Tracker 1110

The clip design above demonstrates the mounting mechanism for the tracker.

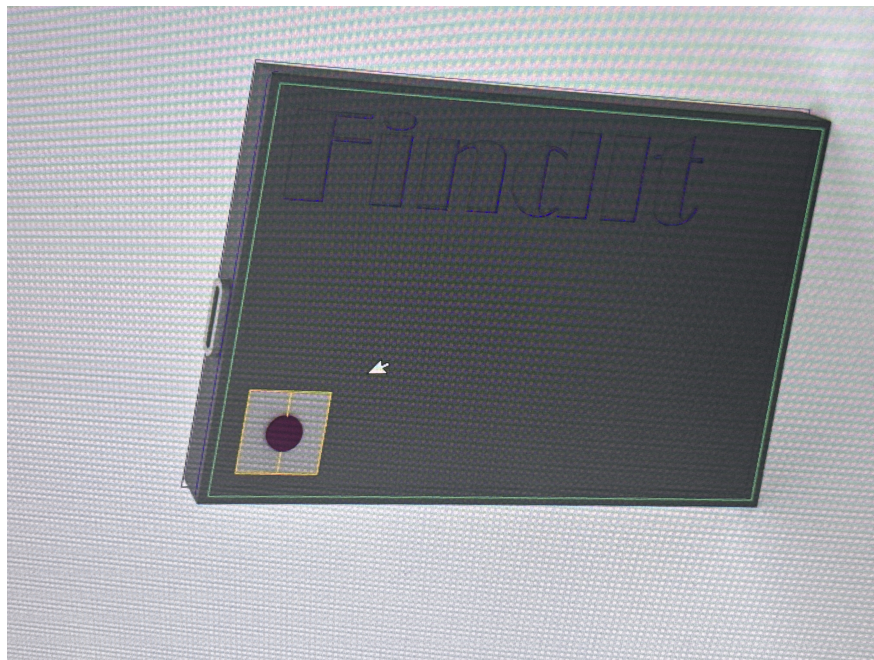


Figure A.6: CaseIsometric: Isometric view of the initial case prototype

The isometric view above provides a full perspective of the initial enclosure prototype.

- Flashed board with default firmware to connect to SenseCraft UI and retrieve LoRaWAN device keys.

A.2.2 Next Week's Plan

- Continue configuring the board to connect to the LoRaWAN network server.
- Implement OpenStreetMaps API in the React frontend for mapping functionality.

A.3 Week Report 12 (12/05/2025)

A.3.1 Progress Made

- Received the Wio Tracker 1110 Development Board.
- Configured The Things Stack LoRaWAN network.
- Initialized database tables and schema.

A.3.2 Next Week's Plan

- Configure the development board with SenseCap to collect device keys.
- Debug the firmware to optimize GPS collection and confirm LoRaWAN uplinks.
- Prepare the React user interface for prototype demonstration.

A.4 Week Report 11 (11/14/2025)

A.4.1 Progress Made

- Created a [Paper Prototype](#) for the FindIt user interface.

- Developed user personas for the [User Interface Design](#).
- Began composing UML diagrams for the [logical view](#) (class diagram), [development view](#) (component diagram), [process view](#) (sequence diagram), and [physical view](#) (package diagram).

A.4.2 Next Week's Plan

- Begin firmware development for the Wio Tracker 1110.
- Finalize UML diagrams for all system views.

A.5 Week Report 10 (11/7/2025)

A.5.1 Progress Made

- Initialized The Things Stack LoRaWAN network in preparation for the development board.
- Reordered the board from a different vendor to reduce wait times.

A.5.2 Next Week's Plan

- Begin work on Preliminary Design: User personas, UI design, Architecture, Use Cases, Logical view, Development view, Process view, Physical view.
- Continue developing the frontend regardless of board arrival.
- Collect board measurements to design the device case.

A.6 Week Report 9 (10/31/2025)

A.6.1 Progress Made

- Created activity diagrams for all six use cases.

- Initialized The Things Stack LoRaWAN server in preparation for the development board.

A.6.2 Next Week's Plan

- Refine requirements and use cases.
- Create system classes that correspond to activity diagrams.

A.7 Week Report 8 (10/24/2025)

A.7.1 Progress Made

- Initialized infrastructure: Supabase database, React frontend, Postman API testing environment, and CI pipeline.
- Populated Kanban board for development tasks.
- Added more activity diagrams to reflect system behaviors.

A.7.2 Next Week's Plan

- Begin UI construction and GPS coordinate testing.
- Finalize requirements, use cases, and user stories.
- Prepare and present the Project Concept Development and Specification Demo.

A.8 Week Report 7 (10/17/2025)

A.8.1 Progress Made

- Created Use Case Diagram for the FindIt system.

- Created Activity Diagram.
- Completed budget update to order the Wio Tracker 1110 development board.

A.8.2 Next Week's Plan

- Begin Arduino firmware development (pending board arrival).
- Begin setting up CI/CD infrastructure and Supabase database.

A.9 Week Report 6 (10/10/2025)

A.9.1 Progress Made

- Updated [Requirements](#) chapter.
- Added [Use Cases](#) chapter.
- Added [User Stories](#) chapter.
- Added Glossary chapter.
- Updated budget for new development board.

A.9.2 Next Week's Plan

- Develop system UML diagrams.
- Meet with Dr. Damiano Zanotto.
- Continue building CI/CD infrastructure and Supabase.
- Await and debug new development board.

A.10 Week Report 5 (10/02/2025)

A.10.1 Progress Made

This week, the team completed and presented our Demo for Project Mission and Development Plan. We set up our GitHub repository and created a project Kanban board to track tasks in an Agile workflow.

A.10.2 Next Week's Plan

Next week, we plan to order the new development board, begin frontend development, configure the Supabase backend, integrate CircleCI, and build our DevOps infrastructure.

A.11 Week Report 4 (09/25/2025)

A.11.1 Progress Made

Our team determined that our current board (incorrect bootloader, XIAO preset) must be replaced with a proper Wio Tracker 1110 to regain full access to GPS, LoRa, and serial interfaces.

A.11.2 Next Week's Plan

We plan to complete Presentation 1, acquire the new board, set up GitHub + Agile boards, and begin GPS and data pipeline development.

A.11.3 Risk Mitigation

We aim to reduce the risk of inaccurate GPS tracking by implementing both indoor and outdoor tracking strategies.

A.12 Week Report 3 (09/18/2025)

A.12.1 Progress Made

The team completed the [Team Declaration](#), defining mission, drivers, and constraints.

A.12.2 Next Week's Plan

We will debug the Sseed LoRaWAN Arduino device: bootloader, buttons, GPS, LoRa module, and serial interface.

A.12.3 Risk Mitigation

The biggest risk is the malfunctioning Sseed LoRaWAN device. Mitigation includes testing alternative hardware such as the Raspberry Pi Nano.

A.13 Week Report 2 (09/12/2025)

A.13.1 Progress Made

We created a system deployment diagram and met with Professor Zanotto to refine system scope toward an autism/dementia tracker leveraging LoRaWAN and local hosting with The Things Stack.

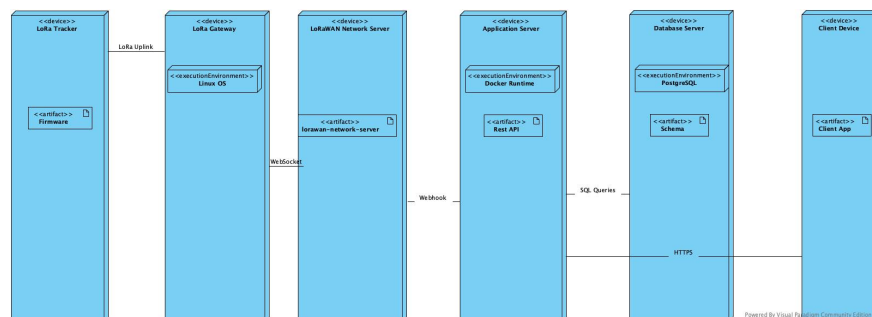


Figure A.7: System Diagram

A.13.2 Next Week's Plan

Ryan will research open-source tracker designs; Roy will research scalability; Cory will research AWS-free workflows. We will continue debugging the LoRaWAN board.

A.14 Week Report 1 (09/05/2025)

A.14.1 Progress Made

Ryan and Cory explored project ideas, consulted classmates and professors, and planned a decision timeline.

A.14.2 Next Week's Plan

Ryan will request continuation of his summer research project on LoRaWAN-based GPS tracking.

A.14.3 Backup Plan

If needed, the team may pivot to a “magic mirror” dashboard project driven by a Raspberry Pi.

Appendix B

Development Notes

- Ryan Davis, Cory Vitanza, and Roy Tas

B.1 Wio Tracker 1110 configs

- Boards manager URL for ArduinoIDE: https://files.seeedstudio.com/arduino/package_seeeduino_boards_index.json
- Boards file: Seeed nRF52 Boards
- Seeed Wio Tracker 1110

B.2 Connecting to Sensecap

- <https://sensecap.seeed.cc/portal/#/dashboard>
- JoinEUI=AppEUI
- Sensecraft Configs:
 - **Platform:** SenseCAP for The Things Network
 - **Frequency Plan:** US915

- Reset board to produce serial monitor output
- Must change frequency to US915 and send to device, then reset board to work.

B.3 Dev Board Boot Configuration Changes

To eliminate the need for configuring the device through the SenseCraft Mate app, the LoRaWAN modem region was set to US915 by default (the firmware defaulted to EU868).

- Edited the following file:

```
~/Library/Arduino15/packages/Seeeduno/hardware/nrf52/1.1.12/libraries/  
WM1110_At_Config/src/WM1110_Param_Var/WM1110_Param_Var.c
```

- Updated `app_param.lora_info.ActiveRegion` from 5 (EU868) to 8 (US915)
- Reflashed the board with the updated firmware

B.4 Sensecap Dashboard API to Database

Sensor data ID's:

- Longitude 4197
- Latitude 4198
- Accel X 4150
- Accel Y 4151
- Accel Z 4152

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